Gauge Fields as Forced Rule-Type Structure: The Closure of Arc Q and Theorem 17

Allen Proxmire Claude (AI collaborator)

April 2026

Abstract

We close Arc Q of the Event-Density (ED) program by promoting the Gauge-field-as-Rule-Type Hypothesis (GRH) from form-level structural commitment to a FORCED-unconditional structural theorem (Theorem 17). Earlier Arc Q work established GRH at the form level (Theorem 5), canonical (anti-)commutation (Theorem 6), and primitive-level UV-finiteness UV-FIN (Theorem 7), but left five refinement axes unresolved: lightlike worldline structure (R-1), gauge-quotient individuation (R-2), commitment-axis structure (R-3), non-Abelian extension (R-4), and the vacuum-vs-particle vacuum-side classification (R-5). We present a four-substage closure trajectory — Q.2 (gauge group + quotient), Q.3 (vertex taxonomy + minimal coupling), Q.7 (worldline + second-quantisation), Q.8 (vacuum + zero-point) — followed by a synthesis stage (global integration + falsifier audit + Theorem 17 statement). All seven refinement items close substage-CANDIDATE-FORCED with all twelve substage-level falsifiers and all six synthesis-level falsifiers dispatched dark; in particular, the Arc M circularity check (SFal-5) verifies that the Theorem 17 derivation chain invokes no downstream input. **Theorem 17 (GRH forced-unconditional)** asserts that the gauge field A_μ is the participation measure of a structural rule-type τ_g whose group / vertex / worldline / vacuum content is forced across all four channels under a single unified gauge-quotient identification, with form FORCED by ED primitives plus Theorems 1–16 and numerical magnitudes (specific gauge group, coupling values, V1 kernel parameters, vacuum energy density) inherited from the value layer. Theorem 17 advances the FORCED-theorem inventory from sixteen to seventeen, closes Arc Q, and unblocks the Arc M F-M8 cascade.

1 Introduction

1.1 Motivation: gauge invariance as structural consequence

In standard quantum field theory, gauge invariance is *postulated*. One selects a Lie group, declares a gauge field with prescribed transformation behaviour, and constructs the matter sector in representations of the group. The procedure is operationally successful but structurally opaque: nothing in the foundational machinery *forces* the existence of a gauge field, *forces* its non-Abelian capability, or *forces* its vacuum structure to take any particular form. Each commitment is a modelling choice.

Event-Density (ED) takes the opposite stance. The ED programme begins from a small primitive stack (the thirteen primitives P-01 through P-13) on a 3+1-dimensional event manifold, and asks which physical structures are *forced* by those primitives plus the consequences derived from them. A "FORCED-unconditional" structural theorem in ED is one whose statement is derivable from primitives and prior FORCED theorems alone, without introducing any new CANDIDATE assumption.

Arc Q opens the question whether *gauge invariance itself* — the existence and structure of the gauge field A_μ — is FORCED in this sense.

1.2 Historical framing: from postulate to derived property

Earlier Arc Q work established three foundational FORCED theorems:

- **Theorem 5** — GRH unconditional at the *form level*: the gauge field is a participation measure of a rule-type τ_g.
- **Theorem 6** — canonical (anti-)commutation relations FORCED.
- **Theorem 7** — primitive-level UV-finiteness (UV-FIN) FORCED.

These results established the form-level identification of the gauge field with a rule-type, but five refinement items remained open: R-1 lightlike worldline; R-2 gauge-quotient individuation; R-3 commitment-axis structure; R-4 non-Abelian extension; R-5 vacuum-vs-particle classification.

1.3 Arc Q's role in the ED programme

Within the broader ED programme, Arc Q occupies the gauge-and-vacuum position in the Phase-2 form-level sector: alongside Arc R (relativistic kinematics), Arc M (chain-mass), and Arc N (kernel-level memory structure), Arc Q owns the gauge-field structural sector.

1.4 Summary of results

This paper presents **Theorem 17 (GRH forced-unconditional)**: the seventeenth FORCED structural theorem in the ED inventory. The result is reached via a four-substage closure trajectory followed by a synthesis stage:

```
Q.1 (form-level baseline, Theorem 5) ---+
                                        |
Q.2 (gauge-group + quotient)            |
Q.3 (vertex + minimal coupling)         |--> Synthesis --> Theorem 17
Q.7 (worldline + second-quantisation)   |    (Memos 00-02)
Q.8 (vacuum + zero-point)               |
                                        +---> Arc M F-M8 cascade
```

All twelve substage-level falsifiers and all six synthesis-level falsifiers were dispatched without trigger. The refinement-closure map advances to 7/7.

2 Background and Preliminaries

2.1 ED primitives load-bearing in Arc Q

The seven primitives load-bearing in Arc Q's closure are:

Primitive	Statement
P-02	Rule-types are structural carriers with participation measures.
P-04	Bandwidth additivity over independent rule-type sectors.
P-06	Commitment events are the fundamental ontology.
P-07	Commitments require an anchor: vertex-anchored at excitation.
P-10	Gauge-quotient identification at primitive level.
P-11	UV-finiteness of structural envelopes.
P-13	Continuous-time evolution closure.

2.2 Prior forced theorems 1–16

Most directly load-bearing for Arc Q's closure: Theorem 5 (GRH form-level), Theorem 6 (canonical (anti-)commutation), Theorem 7 (UV-FIN), Theorem 8 (V1 finite-width vacuum kernel from Arc N), Theorem 16 (time-translation Schrödinger evolution from Phase-1 Arc U3).

2.3 Refinement structure (R-1 through R-5)

- **R-1.** Lightlike worldline for τ_g excitations.
- **R-2.** Gauge-quotient individuation across all channels (FULL = partial + completion).
- **R-3.** Commitment-axis structure (FULL = Q.3-side + Q.7-side + Q.8-side).
- **R-4.** Non-Abelian extension capability.
- **R-5.** Vacuum-vs-particle classification (FULL = excitation + vacuum sides).

2.4 Rule-types and participation measures

The ED ontology assigns to each rule-type τ a *participation measure* encoding its contribution to commitment-rate ledgers across the event manifold. GRH asserts A_μ is identified with the participation measure of τ_g.

3 Arc Q Architecture

Arc Q's closure trajectory is a four-substage program followed by a three-memo synthesis. Each substage was structured as Memo 00 (scoping) + Memos 01–02 (substantive derivations) + Memo 03 (verdict + falsifier dispatch).

3.1 Q.2 — Gauge-group admissibility and quotient structure

Refinements owned: R-2 partial, R-4. Q.2 delivers a non-Abelian-capable gauge-group structure for τ_g with Killing-form and Jacobi closure, gauge-quotient identification at the group level, and structure-constant admissibility.

3.2 Q.3 — Vertex taxonomy and minimal coupling

Refinements owned: R-2 completion, R-3 Q.3-side. Q.3 delivers a vertex taxonomy with the minimal-coupling structural vertex, vertex-quotient extension that pulls back the group-quotient through coupling, and vertex-anchored commitment.

3.3 Q.7 — Worldline and second-quantised structure

Refinements owned: R-1, R-3 Q.7-side, R-5 partial. Q.7 delivers the eikonal/null-geodesic limit forcing lightlike worldlines for τ_g excitations, lifts vertex commitments to an ordered worldline ledger $\{v_i\}$, and supplies second-quantised excitation lifting via Theorem 6.

3.4 Q.8 — Vacuum and zero-point structure

Refinements owned: R-3 Q.8-side, R-5 completion. Q.8 delivers a gauge-invariant, UV-finite, V1-form, stationary vacuum kernel for τ_g (VQ1); three vacuum classes (VQ2); strict non-commitment at vacuum + V1-form additive background functional $B[v;\tau_g]$ contributing to vertex-anchored commitment rates (VQ3); full vacuum-sector gauge-quotient certification (VQ4).

3.5 Synthesis — global integration, falsifier audit, Theorem 17

Three memos: Memo 00 (scoping); Memo 01 (global integration; constructs unified four-axis structural object, unified commitment ledger

$$\Gamma_{\text{global}}[v; s] = \Gamma_{\text{excitation}}[v; \tau_g, s] + B[v; \tau_g],$$

unified gauge-quotient ledger; dispatches SFal-1, SFal-2, SFal-3, SFal-4, SFal-6); Memo 02 (Theorem 17 statement, per-clause proof sketch, SFal-5 dispatch, FORCED-unconditional verdict, Arc M cascade map).

4 Statement of Theorem 17 (GRH)

4.1 Physics-facing statement

Theorem 1 (Gauge-field-as-Rule-Type Hypothesis, FORCED-unconditional)**.** *In the Event-Density framework, the gauge field A_μ is the participation measure of a structural rule-type τ_g. The rule-type τ_g admits a non-Abelian-capable gauge-group structure with Killing-form and Jacobi closure; its excitations propagate on lightlike worldlines with second-quantised excitation lifting; commitments are vertex-anchored at the excitation level via a structural minimal-coupling vertex; the vacuum sector carries a gauge-invariant, UV-finite, V1-form, stationary fluctuation envelope, is strict-non-committing, and contributes a gauge-invariant V1-form background functional $B[v; \tau_g]$ additively to vertex-anchored commitment rates in non-vacuum states; all four channels (group, vertex, worldline, vacuum) respect a single unified gauge-quotient identification under the gauge group. The form is forced by ED primitives P-02, P-04, P-06, P-07, P-10, P-11, P-13 and Theorems 1–16; the numerical magnitudes are inherited from the value layer.*

4.2 Structural / refinement-indexed statement

The theorem decomposes into nine clauses (C1)–(C9):

(C1) Carrier. τ_g is a rule-type (P-02) with non-empty participation measure on the gauge-Lie-algebra fibre (Theorem 5 baseline).

(C2) Group structure. Non-Abelian-capable group with Killing-form and Jacobi closure; gauge-equivalent group elements identified (R-2 partial + R-4; Q.2).

(C3) Vertex. Vertex-anchored commitments; minimal-coupling structural vertex; vertex-quotient as pullback of group-quotient (R-2 completion + R-3 Q.3-side; Q.3).

(C4) Worldline. Lightlike worldlines (R-1); vertex ledger lifts to worldline ledger $\{v_i\}$; Fock-style count via Theorem 6 (Q.7).

(C5) Vacuum kernel. Gauge-invariant, UV-finite, V1-form, stationary fluctuation envelope (R-5 completion form-side; Q.8 Memo 01; Theorems 7, 8).

(C6) Vacuum classification. Three classes (background / excitation / mixed); excitation-vacuum minimal FORCED (Q.8 Memo 01).

(C7) Vacuum commitment. Strict non-commitment + additive $B[v; \tau_g]$:

$$\Gamma_{\text{global}}[v; s] = \Gamma_{\text{excitation}}[v; \tau_g, s] + B[v; \tau_g]$$

in non-vacuum states (R-3 Q.8-side; Q.8 Memo 02; Synthesis Memo 01 §3).

(C8) Unified quotient. All four channels respect a single unified gauge-quotient identification (Q.8 VQ4 + Synthesis Memo 01 §4).

(C9) Acyclic derivability. (C1)–(C8) derivable from primitives + Theorems 1–16 + Q-substage verdicts alone (Synthesis Memo 01 §6.5 + Synthesis Memo 02 §4).

Status: form FORCED, value INHERITED. **forced-unconditional at the structural level.**

5 Proof Sketch of Theorem 17

The proof traces each clause to its primitive-level, theorem-level, and Q-substage provenance.

5.1 (C1) Carrier

Provenance: P-02 + GRH-1 + GRH-2 + Theorem 5. Form-level identification established by Theorem 5 baseline. **No additional candidate.**

5.2 (C2) Group structure

Provenance: P-10 + Q.2 verdict + R-2 partial + R-4. Q.2 Memo 01 closes R-4 via Killing form, Jacobi identity, and structure-constant admissibility. Q.2 Memo 02 closes R-2 partial via the gauge-quotient extension at group level.

5.3 (C3) Vertex

Provenance: P-06 + P-07 + P-10 + Q.3 verdict + R-2 completion + R-3 Q.3-side. Q.3 Memo 01 derives the vertex taxonomy and the minimal-coupling structural vertex; Q.3 Memo 02 closes R-2 completion and R-3 Q.3-side.

5.4 (C4) Worldline

Provenance: P-13 + Q.7 verdict + Theorem 6 + R-1 + R-3 Q.7-side + R-5 partial. Q.7 Memo 01 closes R-1 via the eikonal/null-geodesic limit; Q.7 Memo 02 lifts the vertex ledger and supplies second-quantised excitation lifting via Theorem 6.

5.5 (C5) Vacuum kernel

Provenance: P-02 + P-04 + P-10 + P-11 + P-13 + Theorem 7 + Theorem 8 + Q.8 Memo 01. Six constraints: non-vacuity (P-02 + GRH-2); UV-finiteness (P-11 + Theorem 7); V1 form (Theorem 8); gauge invariance (R-2 + P-10); bandwidth additivity (P-04); stationarity (P-13 + Theorem 16).

5.6 (C6) Vacuum classification

Provenance: Q.7 worldline + VQ1 + R-2 + Q.8 Memo 01. Two orthogonal axes (background expectation × worldline support) yield three admissible classes; excitation-vacuum minimal FORCED.

5.7 (C7) Vacuum commitment

Provenance: P-04 + P-06 + P-07 + Q.8 Memo 02 + Synthesis Memo 01 §3. P-07 forces strict non-commitment at vacuum; P-04 forces additive composition; $B[v; \tau_g]$ inherits from VQ1.

5.8 (C8) Unified quotient

Provenance: P-10 + Q.8 Memo 02 + Synthesis Memo 01 §4 + R-2 + R-4. Synthesis Memo 01 §4 dispatches the global six-pair cross-channel audit; all PASS.

5.9 (C9) Acyclic derivability

Provenance: Synthesis Memo 02 §4. Per-clause trace verifies that every input traces to a primitive, a Theorem 1–16, or a Q-substage verdict. F-M8 is a *consequence* of Theorem 17, not input. **SFal-5 NOT TRIGGERED.**

5.10 Falsifier-level provenance

Falsifier source	Status
Q.2 substage F-falsifiers	All NOT TRIGGERED
Q.3 substage V-falsifiers	All NOT TRIGGERED
Q.7 substage W-falsifiers (incl. WFal-5 circularity)	All NOT TRIGGERED
Q.8 substage VQ-falsifiers (incl. VQFal-4 circularity)	All NOT TRIGGERED
Synthesis SFal-1 (cross-substage)	NOT TRIGGERED
Synthesis SFal-2 (global quotient)	NOT TRIGGERED
Synthesis SFal-3 (commitment-ledger)	NOT TRIGGERED
Synthesis SFal-4 (primitive global)	NOT TRIGGERED
Synthesis SFal-5 (Arc M circularity)	NOT TRIGGERED
Synthesis SFal-6 (acyclicity)	NOT TRIGGERED

All eighteen falsifiers dispatched dark. **No falsifier blocks Theorem 17.** □

6 Structural Consequences

6.1 Gauge invariance as interface property of τ_g

Gauge invariance is the *interface property* of τ_g: the equivalence relation under which structurally-identical τ_g configurations are identified at primitive level (P-10). Transformation freedom is not a redundancy; it is the absence of distinction between structurally-identical primitive-level objects.

6.2 Masslessness as structural, not contingent

Form-level masslessness (C4) is FORCED; value-level mass corrections are INHERITED via Arc M's chain-mass ledger (F-M8 cascade). No Higgs mechanism is invoked or excluded.

6.3 Unified quotient structure

Clause (C8) certifies a single gauge-quotient identification across all four channels. Channel-specific gauge fixings are structurally inadmissible.

6.4 Vacuum-level commitment rule

The structural form of the commitment-rate ledger:

$$\Gamma_{\text{global}}[v; s] = \Gamma_{\text{excitation}}[v; \tau_g, s] + B[v; \tau_g].$$

The vacuum is strict-non-committing in pure vacuum states; in non-vacuum states the vacuum kernel contributes additively.

6.5 Worldline / vacuum / vertex consistency

The four-axis × three-branch × four-channel global object is internally consistent. The worldline degenerates cleanly into the vacuum; the vacuum kernel feeds back into the vertex commitment rate; the vertex commits anchor the worldline ledger.

7 Implications for Arc M

7.1 F-M8 promotion

Theorem 17 satisfies F-M8's antecedent via clauses (C5), (C7), (C8), (C9). **F-M8 promotes from candidate to forced-unconditional within Arc M cascade scope.**

7.2 Massless-slot resolution

M.1 will resolve whether τ_g itself acquires mass through the F-M8 channel. Anticipated answer: τ_g remains massless at form level; F-M8 fires on other rule-types via cross-sector coupling.

7.3 Cross-sector coupling constraints

The unified quotient (C8) constrains how $B[v; \tau_g]$ couples to other rule-types' commitment rates. M.3 will derive the structural coupling rule.

7.4 Downstream M.1–M.4 cascade

M.1 (massless-slot resolution), M.2 (mass-form additivity), M.3 (τ_g cross-sector coupling), M.4 (Arc M synthesis verdict refresh). Acyclicity preserved.

8 Discussion

8.1 Conceptual shift: gauge fields as rule-types

Theorem 17 reframes the gauge field as a derived structural object. The form-level content is forced from the primitive layer; the numerical value layer remains inherited.

8.2 Comparison with traditional gauge theory

Traditional QFT	**Event-Density (Theorem 17)**
Gauge group postulated	Non-Abelian-capable structure FORCED (C2).
A_μ postulated	Participation measure of τ_g (C1).
Minimal coupling postulated	FORCED as structural vertex (C3).
Vacuum via path integral	Vacuum kernel FORCED as gauge-inv. V1 form (C5–C7).
Gauge invariance postulated	Unified gauge-quotient as primitive identification (C8).
Higgs mechanism postulated	Form masslessness (C4) + value-layer F-M8 (Arc M).

The shift is ontological: gauge structure is *derived*, not *imposed*.

8.3 Implications for unification and emergent structure

A single unified gauge-quotient (C8) across all four channels: structural unification at the rule-type level. Empirical-unification implications are value-layer questions.

8.4 Limitations and open questions

Theorem 17 does *not* derive: the specific gauge group of nature, specific coupling magnitudes, representation content, the Higgs mechanism, vacuum-energy magnitudes, anomaly cancellation, or RG running. Open questions: M.1–M.4 cascade closure; Phase-3 GR integration of $B[v; \tau_g]$; empirical signatures; multi-rule-type gauge sectors.

9 Conclusion

Arc Q closes via a four-substage trajectory (Q.2, Q.3, Q.7, Q.8) plus a three-memo synthesis. All seven refinement items close at substage level with all eighteen falsifiers dispatched dark. The synthesis stage promotes GRH from form-level (Theorem 5) to FORCED-unconditional (Theorem 17), the seventeenth FORCED structural theorem in the ED inventory.

Theorem 17 supplies the structural foundation for ED's gauge-and-vacuum sector, completing the Phase-2 form-level closure (with Arc R, Arc M, Arc Q) plus the kernel-level closure (Arc N) plus the Phase-1 emergence layer. The immediate downstream consequence is the Arc M F-M8 cascade; beyond that, Phase-3 GR inherits $B[v; \tau_g]$ as a candidate channel for vacuum-mediated curvature corrections.

A Detailed Refinement-Closure Map

Refinement item	**Status**	**Closing substage**
R-1 (lightlike worldline)	CLOSED	Q.7 (Memo 01, W1+W2)
R-2 partial (group-level quotient)	CLOSED	Q.2 (Memo 02, F3)
R-2 completion (vertex-quotient)	CLOSED	Q.3 (Memo 02, V3+V4)
R-2 FULL	CLOSED	Q.2 + Q.3
R-3 Q.3-side (vertex commitment)	CLOSED	Q.3 (Memo 02)
R-3 Q.7-side (worldline commitment)	CLOSED	Q.7 (Memo 02, W3+W4)
R-3 Q.8-side (vacuum commitment)	CLOSED	Q.8 (Memo 02, VQ3)
R-3 FULL	CLOSED	Q.3 + Q.7 + Q.8
R-4 (non-Abelian extension)	CLOSED	Q.2 (Memo 01, F2)
R-5 partial (excitation-side)	CLOSED	Q.7 (Memo 02, W3+W4)
R-5 completion (vacuum-side)	CLOSED	Q.8 (Memos 01 + 02)
R-5 FULL	CLOSED	Q.7 + Q.8

Map: 7/7 refinement items CLOSED; 5/5 top-level refinement axes FULLY CLOSED.

B Primitive-Level Audits

B.1 Per-substage primitive load

Primitive	**Q.2**	**Q.3**	**Q.7**	**Q.8**	**Synth.**	**Forbidden?**
P-02	supports	supports	supports	constrains	supports	No.
P-04	neutral	constrains	constrains	constrains	constrains	No.
P-06	neutral	constrains	constrains	constrains	constrains	No.
P-07	neutral	constrains	constrains	constrains (decisive)	constrains	No.
P-10	constrains	constrains	constrains	constrains	constrains	No.
P-11	neutral	neutral	neutral	constrains	constrains	No.
P-13	neutral	neutral	constrains	constrains	constrains	No.

No primitive forbids any Arc Q result; no new CANDIDATE introduced.

C Falsifier Tables

C.1 Substage falsifier ledger

Falsifier set	Substage	Status
F-1 through F-5	Q.2	All NOT TRIGGERED
V-1 through V-5	Q.3	All NOT TRIGGERED
WFal-1 through WFal-8 (incl. WFal-5)	Q.7	All NOT TRIGGERED
VQFal-1 through VQFal-7 (incl. VQFal-4)	Q.8	All NOT TRIGGERED

C.2 Synthesis falsifier ledger

Falsifier	Tested in	Status
SFal-1 (cross-substage)	Synth. Memo 01 §6.1	NOT TRIGGERED
SFal-2 (global quotient)	Synth. Memo 01 §6.2	NOT TRIGGERED
SFal-3 (commitment-ledger)	Synth. Memo 01 §6.3	NOT TRIGGERED
SFal-4 (primitive global)	Synth. Memo 01 §6.4	NOT TRIGGERED
SFal-5 (Arc M circularity)	Synth. Memo 02 §4	NOT TRIGGERED
SFal-6 (acyclicity)	Synth. Memo 01 §6.5	NOT TRIGGERED

Cumulative: 18 falsifiers, all NOT TRIGGERED.

D Worldline / Field Correspondence Derivation

The lightlike-worldline structure for τ_g excitations (clause C4 / R-1) is derived in Q.7 Memo 01 via the eikonal limit:

1. By P-13 and Theorem 16, every rule-type carrier admits a continuous-time evolution generator.
2. By the form-level vanishing of structural mass-form for τ_g, the eikonal-limit dispersion is null: $g^{\mu\nu}k_\mu k_\nu = 0$.
3. The associated worldline is a null geodesic: $g^{\mu\nu}\dot{x}_\mu \dot{x}_\nu = 0$.
4. The vertex-anchored commitment ledger lifts onto the null worldline as ordered events $\{v_i\}$ along the affine parameter.
5. Second-quantised excitation lifting inherits Theorem 6, supplying a Fock-style count.

The field-side correspondence is the standard map between worldline ledger and participation measure A_μ.

E Synthesis Falsifier Dispatch Ledger

- **SFal-1:** all six pairs and four triples PASS.
- **SFal-2:** all six cross-channel pairs PASS; non-Abelian global consistency PASSES.
- **SFal-3:** unified rule consistent; no double-counting; acyclicity preserved.
- **SFal-4:** all seven primitives consistent globally.
- **SFal-5:** per-clause trace verifies no Arc M input invoked.

- **SFal-6:** dependency graph has no back-edges.

All six dispatched dark.

References

[1] A. Proxmire, *Event-Density Primitive Stack* (Primitives 01–13), 2026.

[2] A. Proxmire, *Phase-1 Synthesis: Non-Relativistic QM-Emergence*, 2026.

[3] A. Proxmire, *Why the Universe is Structurally Mandatory: Arc R*, 2026.

[4] A. Proxmire, *Chain-Mass Synthesis: Arc M*, 2026.

[5] A. Proxmire, *Non-Markovian Structure as a Forced Memory-Kernel Layer of Event-Density Theory*, 2026.

[6] A. Proxmire, *Quantum Field Theory from Event-Density Primitives: Arc Q form-level baseline*, 2026.

[7] A. Proxmire, *Arc Q Substage Q.2 Memos 00–03*, `arcs/arc-Q/01-04`, 2026.

[8] A. Proxmire, *Arc Q Substage Q.3 Memos 00–03*, `arcs/arc-Q/05-08`, 2026.

[9] A. Proxmire, *Arc Q Substage Q.7 Memos 00–03*, `arcs/arc-Q/09-12`, 2026.

[10] A. Proxmire, *Arc Q Substage Q.8 Memos 00–03*, `arcs/arc-Q/13-16`, 2026.

[11] A. Proxmire, *Arc Q Synthesis Memos 00–02*, `arcs/arc-Q/17-19`, 2026.

[12] A. Proxmire, *Arc M F-M8 Cascade Memo*, `arcs/arc-M/cascade_memo_F_M8_promotion.md`, 2026.

[13] A. Proxmire, *U3 Time-Translation and Schrödinger Evolution* (Theorem 16), 2026.

[14] A. Proxmire, *ED Forced-Theorem Inventory (Theorems 1–17)*, `memory/project_qm_emergence_arc.md`, 2026.

Manuscript closure: 2026-04-27. Companion documents: Arc R paper, Arc M paper, Arc N paper, U3 paper, Phase-1 synthesis.